CHAPTER 2

BASIC COMPUTER NETWORKING

[CO1005 • [Semester] • [Academic year]

Ho Chi Minh University of Technology (HCMUT)
Faculty of Computer Science and Engineering (CSE)
Instructor: [Full name]



For academic purposes only.

Learning Objectives

After this lesson, students will be able to…

1. **Understand Core Concepts of OSI and TCP/IP Models**

   Grasp the definition of the OSI model, reasons for use, and basic layered structure.

2. **Explain the Functions of Network Layers**

   Describe in detail the role of each layer from Physical, MAC, Network to Application.

3. **Apply and Distinguish Network Devices and Protocols**

   Analyze the operating principles of Hub, Switch, Router, and HTTP, TCP/IP protocols.

4. **Identify IoT and Edge Computing Trends**

   Relate foundational networking knowledge to real-world IoT and edge computing applications.

Lesson Roadmap

Structure of Chapter 2: Basic Computer Networking



Foundations

OSI Model Layers

PHY, MAC & Network

Protocols & Devices

TCP/IP & HTTP

Hub, Switch & Router

IoT Ecosystem

RF & Hardware Platforms

Edge Computing & AI

Applications

Smart Agriculture

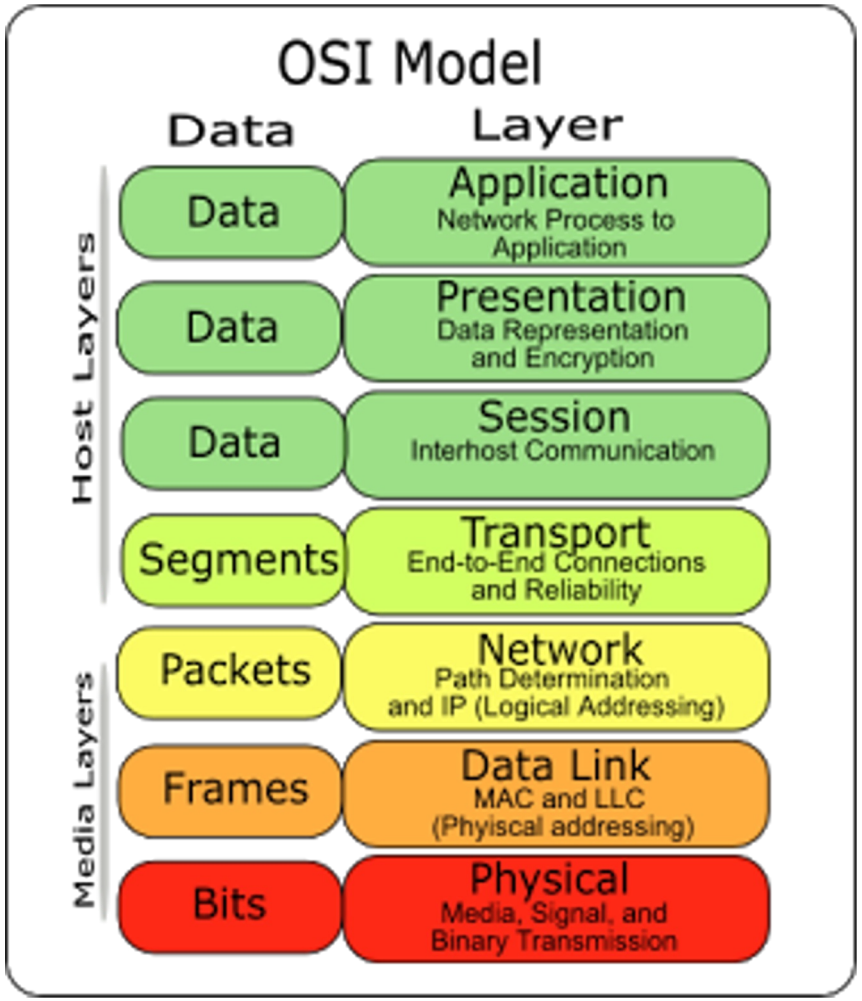
Teachable Machine Demo

01

OSI MODEL & TCP/IP MODEL

Week [x] • CO1005 • [Class] • [Instructor]

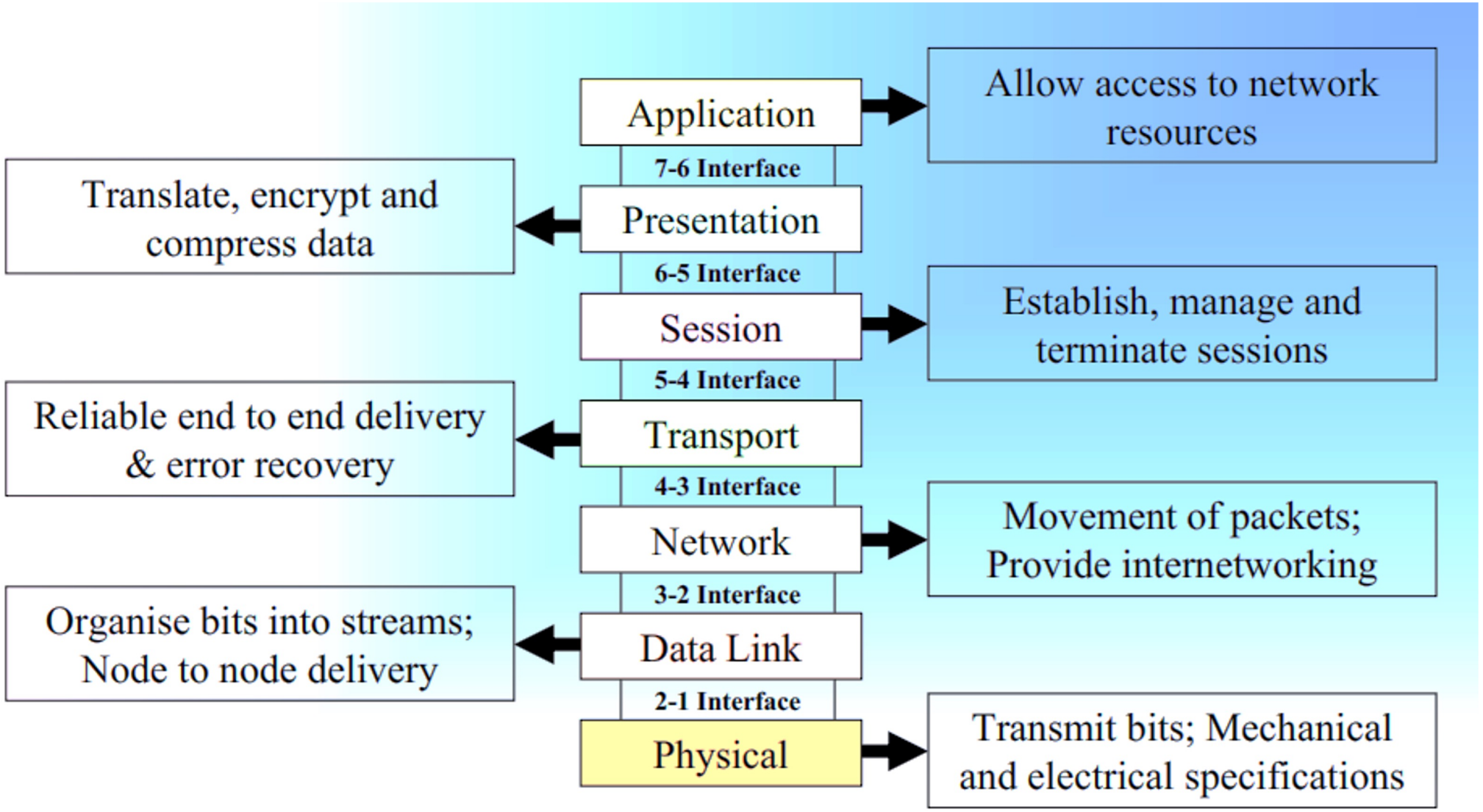
OSI Model



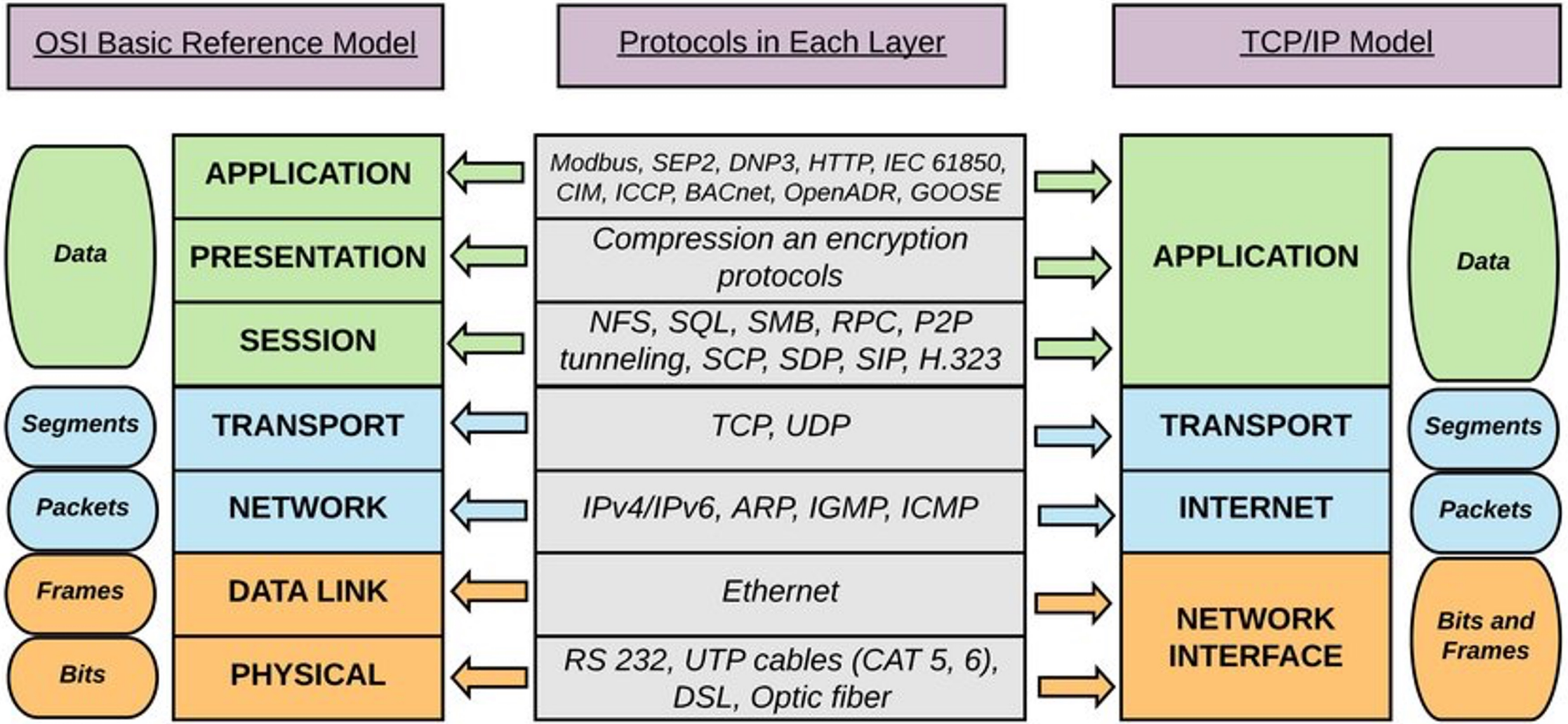
What is OSI model? Why do we have to use OSI model?

- OSI is Open System Interconnection.
- OSI model is used for System and Device Design and Trouble-shooting
- Layers of OSI model:
  - PHY layer
  - MAC layer
  - Network layer
  - Transport
  - Session layer
  - Presentation layer
  - Application layer

OSI Model: Layer Functions



TCP/IP Model compared to OSI Model



02

COMMUNICATION CHANNELS & COMPUTER NETWORKS

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Communication Channel

Definition

- Communication channels carry data from one computer to another.

Physical connections

- Twisted-pair cable consists of pairs of copper wire that are twisted together. Both standard telephone lines and Ethernet cables use twisted pair



- Coaxial cable, a high-frequency transmission cable, replaces the multiple wires of telephone lines with a single solid-copper core



- Fiber-optic cable transmits data as pulses of light through tiny tubes of glass



Communication Channel

Wireless Connections: Wireless connections do not use a solid substance to connect devices. Most use radio waves.

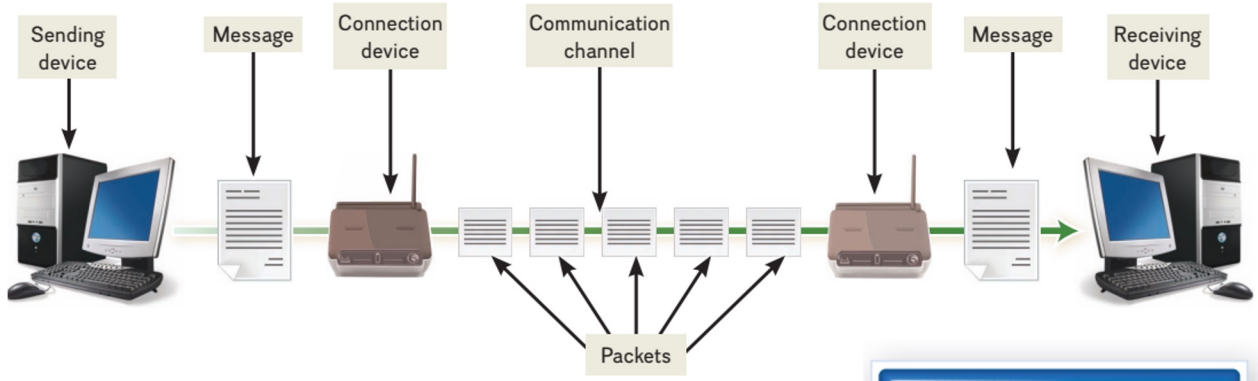
- Bluetooth—transmits data over short distances; widely used for a variety of wireless devices.
- Wi-Fi (wireless fidelity)—uses high-frequency radio signals; most home and business wireless networks use Wi-Fi.
- Microwave—line-of-sight communication; used to send data between buildings; longer distances require microwave stations.
- WiMax (Worldwide Interoperability for Microwave Access)—extends the range of Wi-Fi networks using microwave connections.
- Cellular—uses cell towers to send and receive data within relatively small geographic regions or cells.
- Satellite—uses microwave relay stations; GPS (global positioning system) tracks geographic locations.
- Infrared—uses light waves over a short distance; line-of-sight communication.



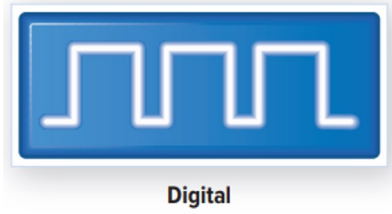
Data Transmission

Data transmitted over a network can exist in the form of analog or digital signals.

Analog: continuous signal. Digital: discrete signal (0 and 1)







01

Bandwidth

measures a communication channel's width or capacity. Four bandwidths are voiceband (low bandwidth), medium band, broadband (high-capacity transmissions), and baseband.

02

Protcols

are rules for exchanging data. Widely used Internet protocols include https and TCP/IP.

03

IP address

(Internet protocol addresses) are unique numeric Internet addresses. DNS (domain name server) converts text-based addresses to and from numeric IP addresses.

03

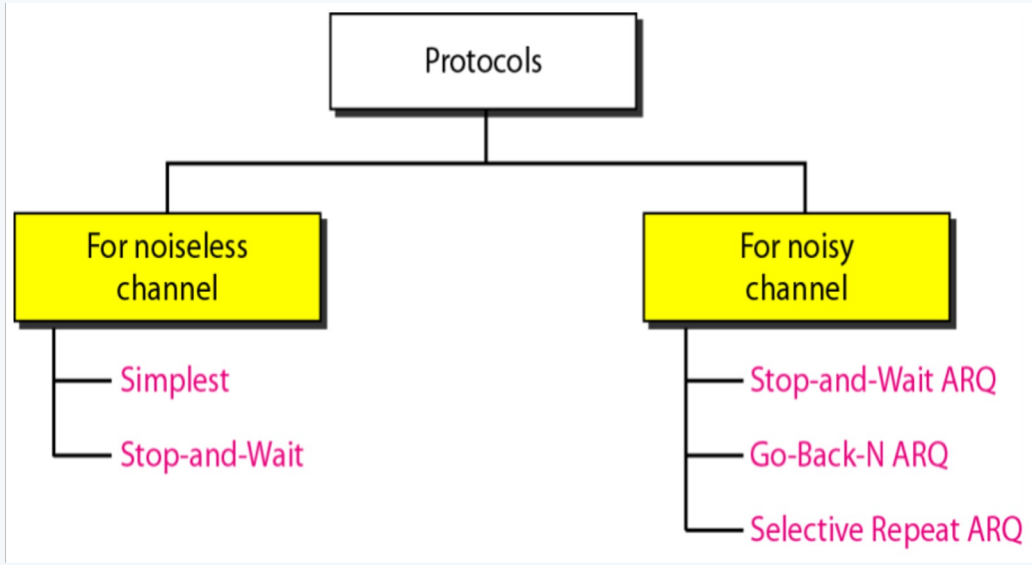
Packets

are small parts of messages.

Communication Protocol

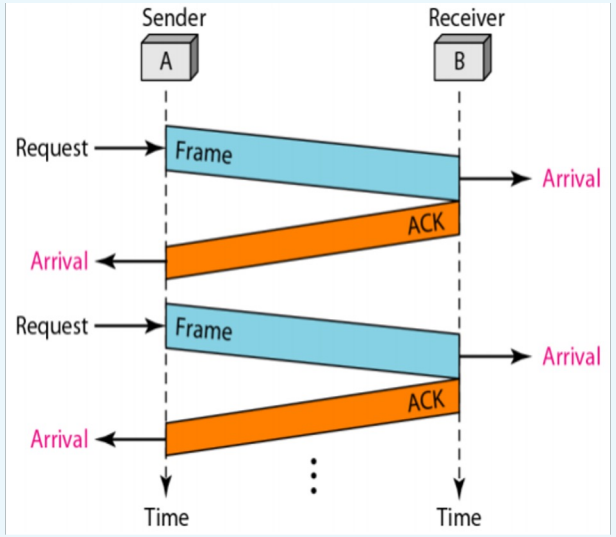
Definition

- Communication protocol is the combination of framing, flow control, and error control to achieve the delivery of data from one node to another.



Implementation

- The protocols are normally implemented in software by using one of the common programming languages.



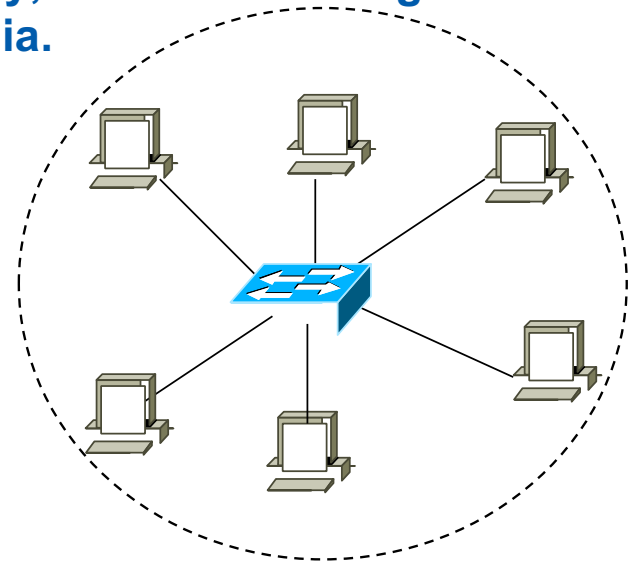
LAN vs. WAN

Local Area Network

LAN

A local area network (LAN) is a computer network that interconnects computers in a limited area such as a home, school, computer laboratory, or office building using network media.

- Medium:
  - Coaxial cable
  - Twisted-pair line

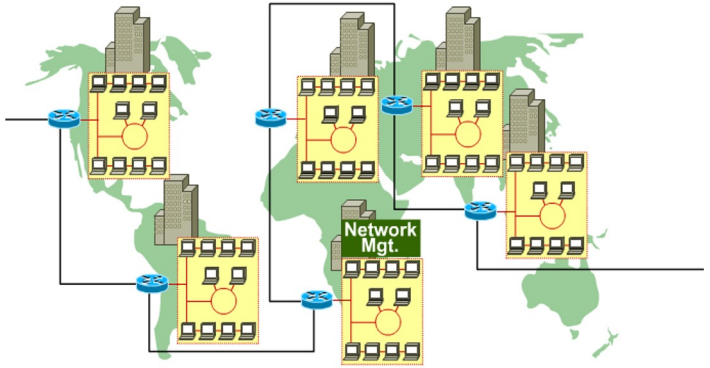


Wide Area Network

WAN

A Wide Area Network (WAN) is a network that covers a broad area (i.e., any telecommunications network that links across metropolitan, regional, or national boundaries) using private or public network transports.

- Besides, several network types:
  - Home
  - WLAN
  - PAN
  - MAN



Network Types

Type	Description
LAN	Local area network; located within close proximity
Home	Local area network for home and apartment use; typically wireless
WLAN	Wireless local area network; all communication passes through access point
PAN	Personal area network; connects digital devices, such as PDAs
MAN	Metropolitan area network; typically spans cities with coverage up to 100 miles
WAN	Wide area network for countrywide or worldwide coverage

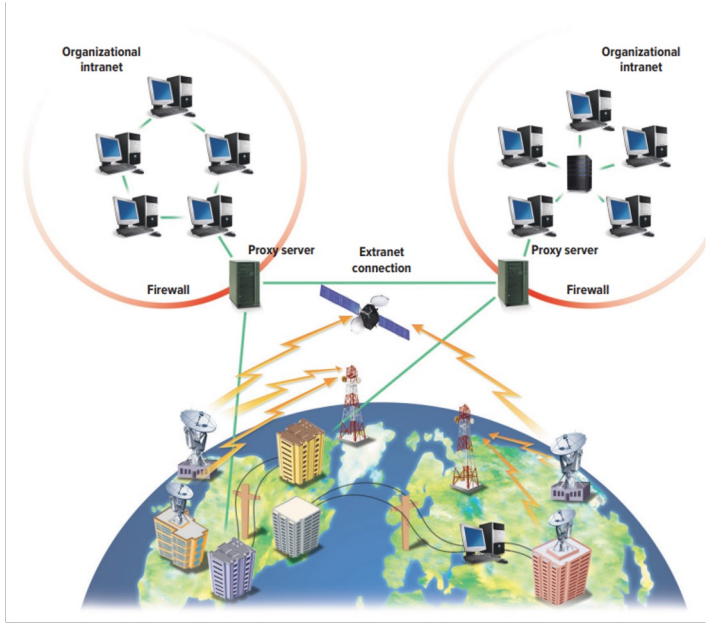
Internet technologies

Intranet

Intranet—private network within an organization; uses browsers, websites, and web pages. Typical applications include electronic telephone directories, e-mail addresses, employee benefit information, internal job openings, and much more.

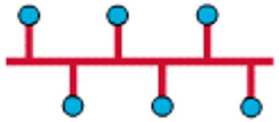
Extranet

Extranet—like an intranet, except it connects more than one organization; typically allows suppliers and others limited access to their networks

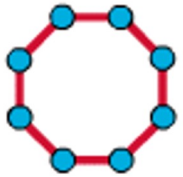


Internet technologies support effective communication using intranets and extranets.

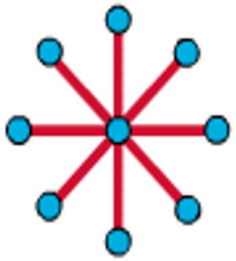
Physical Network Topology



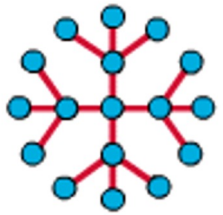
Bus



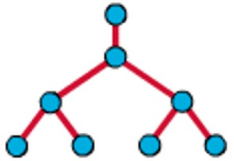
Ring



Star



Extended Star



Hierarchical



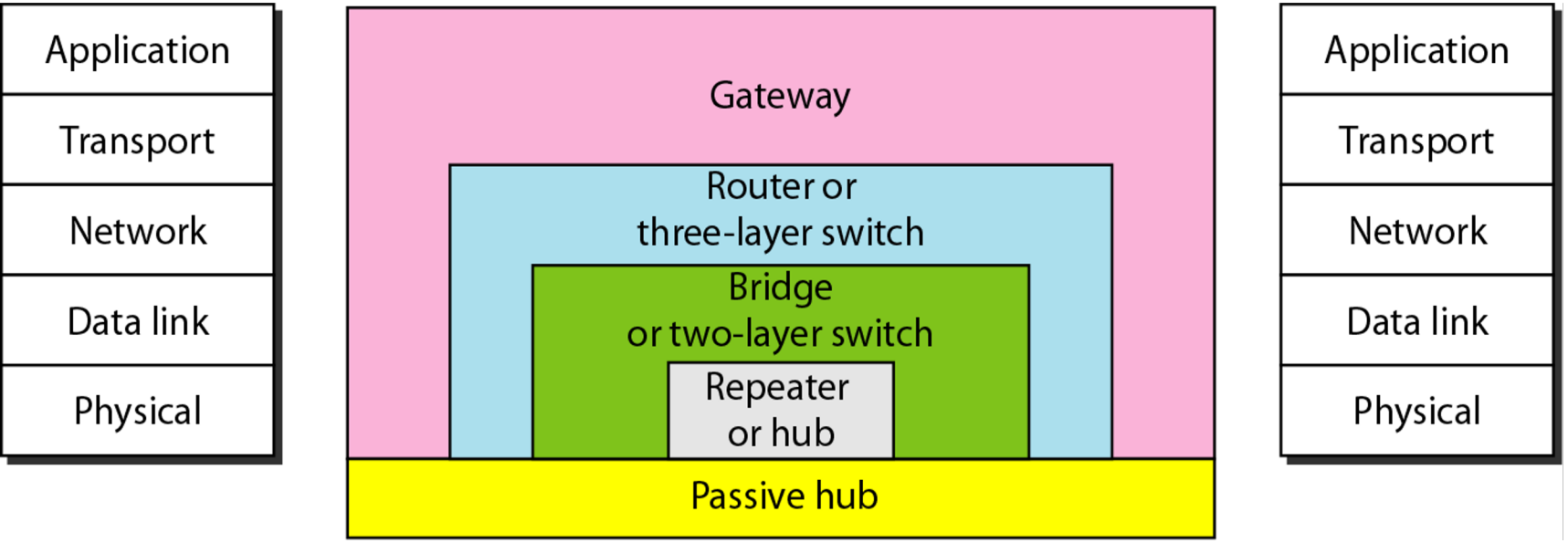
Mesh

03

NETWORK DEVICES & SECURITY

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Network Devices

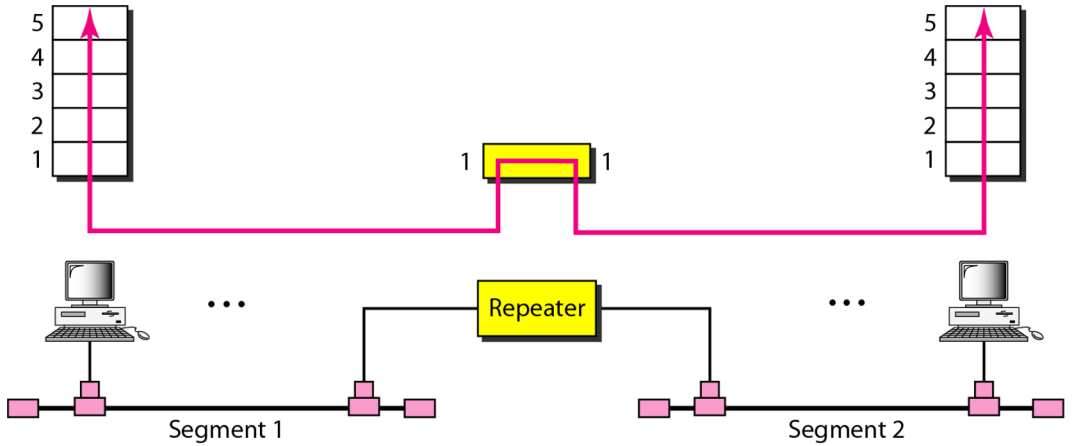


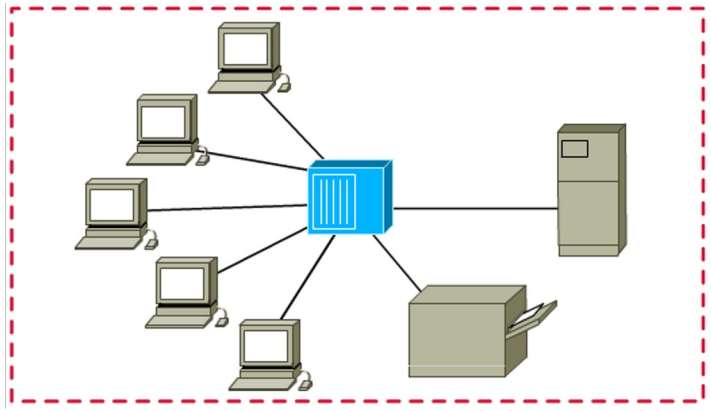
Network devices

- Hub/Repeater
- Switch/Bridge
- Router/3-layer switch
- Gateway
- Firewall

Hub/Repeater

Collision domains are the areas where collisions occur. All of the layer 1 interconnections are part of the collision domain.





01 **Propagate signals from one port to others.**

02 **Layer 1 devices: connect segments of a LAN.**

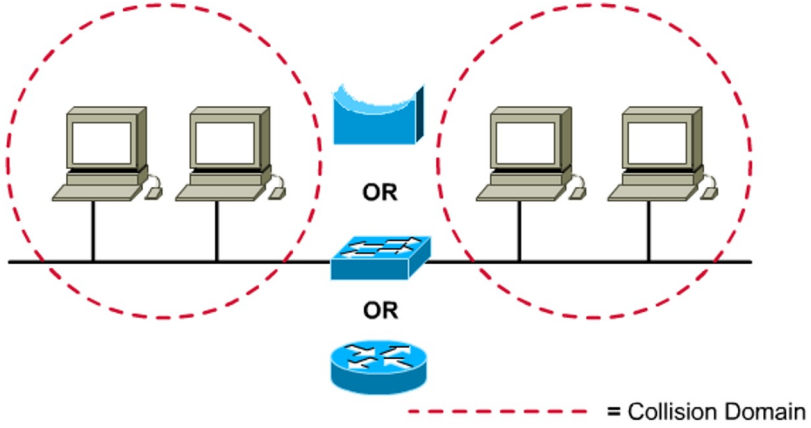
03 **Extending a network with a repeater or a hub results in a larger collision domain.**

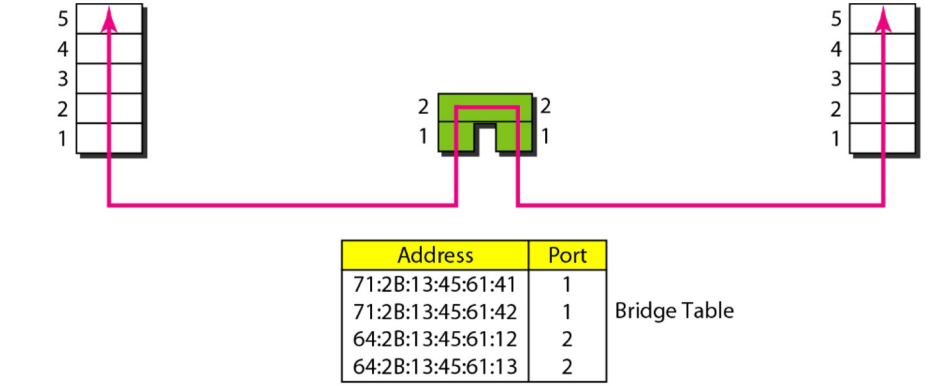
Switch

- A multi-port bridge
- Segmenting Collision Domain
- Cannot prevent layer 2 broadcast domain



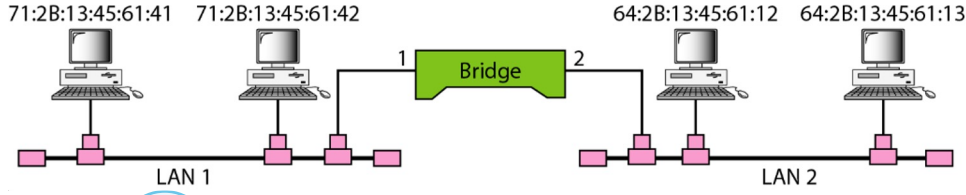
Ethernet Switch





Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table



01 **Support full duplex.**

02 **Hardware switching**

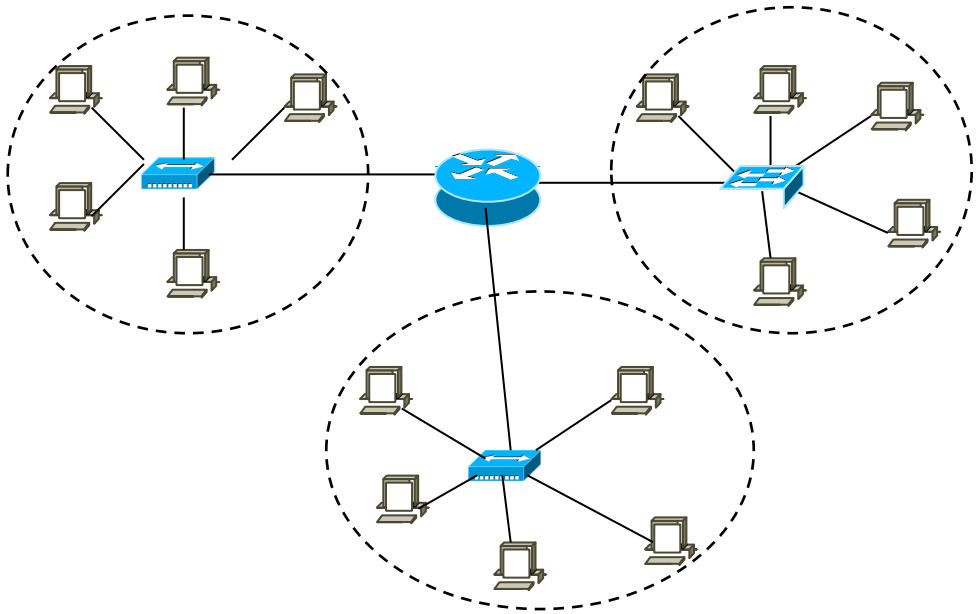
03 **Dynamically builds and maintains a Content-Addressable Memory (CAM) table**

Router

Layer 3 device



Cisco Router



01 **Routing packets to the destination based on the destination IP address**

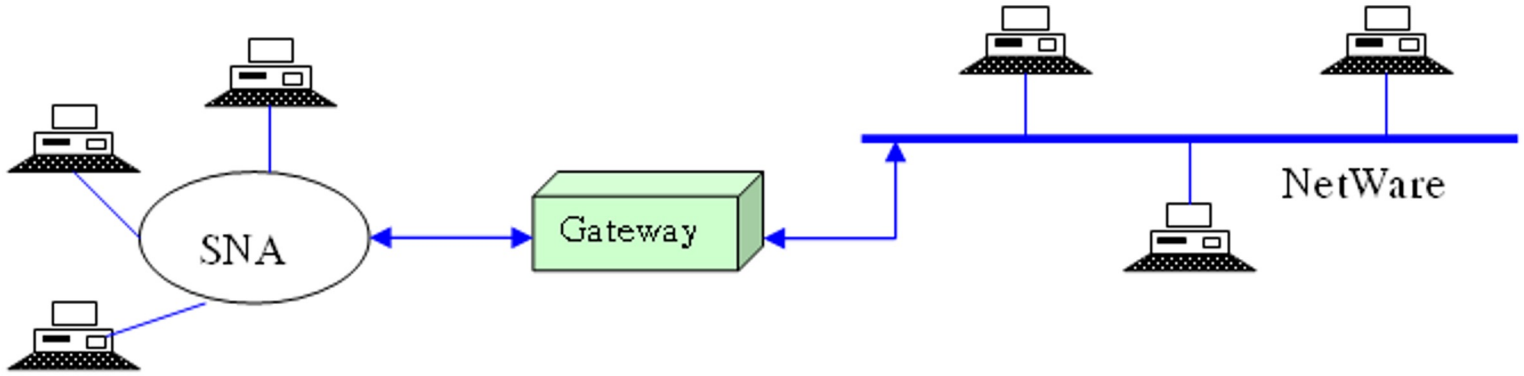
02 **Running routing protocol**

03 **Split layer 2 broadcast domain**

Gateway



Synway E1/T1 Trunk Gateway for VoIP



01 **The gateway acts as a "translator" in between two different systems that use dissimilar communication protocols, data formats, or architectures.**

02 **Operate at the application layer in the OSI model.**

Network Security

Three technologies commonly used to ensure network security are firewalls, intrusion detection systems, and virtual private networks.



Sophos XG 450 Next-Generation Firewall

01

Firewall

controls access; all communications pass through proxy server.

02

Intrusion detection systems (IDS)

Work with firewalls; use sophisticated statistical techniques to recognize and disable network attacks.

03

Virtual private network (VPN)

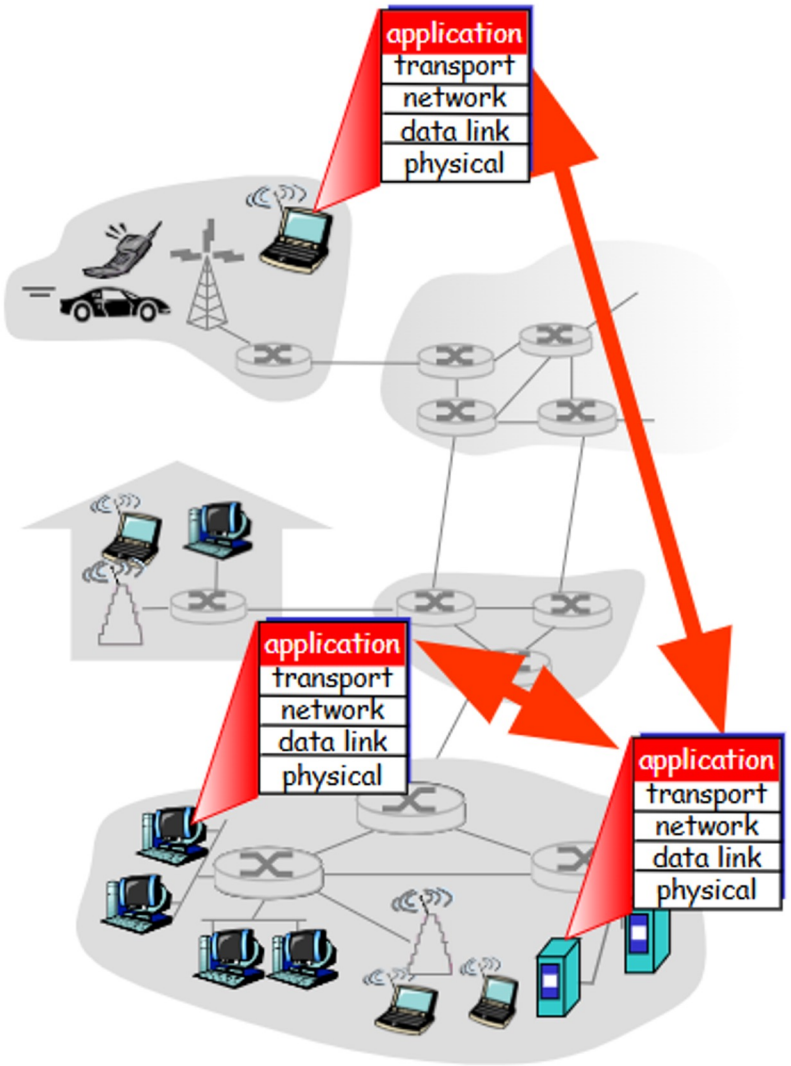
creates a secure private connection between a remote user and the organization's internal network

04

APPLICATION LAYER & WEB PROTOCOLS

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Creating a network app



Write programs that

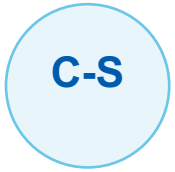
- run on (different) end systems
- communicate over network
- e.g., web server software communicates with browser software

No need to write software for network-core devices.

- Network-core devices do not run user applications
- Applications on end systems allow for rapid app development, propagation

Client-Server vs. P2P

Client-Server



Server

- always-on host
- permanent IP address
- server farms for scaling

Clients

- communicate with server
- may be intermittently connected
- may have dynamic IP addresses
- do not communicate directly with each other

Easier to manage, higher security, but hard to scale

Peer-to-Peer



Peer

- no always-on server
- arbitrary end systems directly communicate.
- peers are intermittently connected and change IP addresses

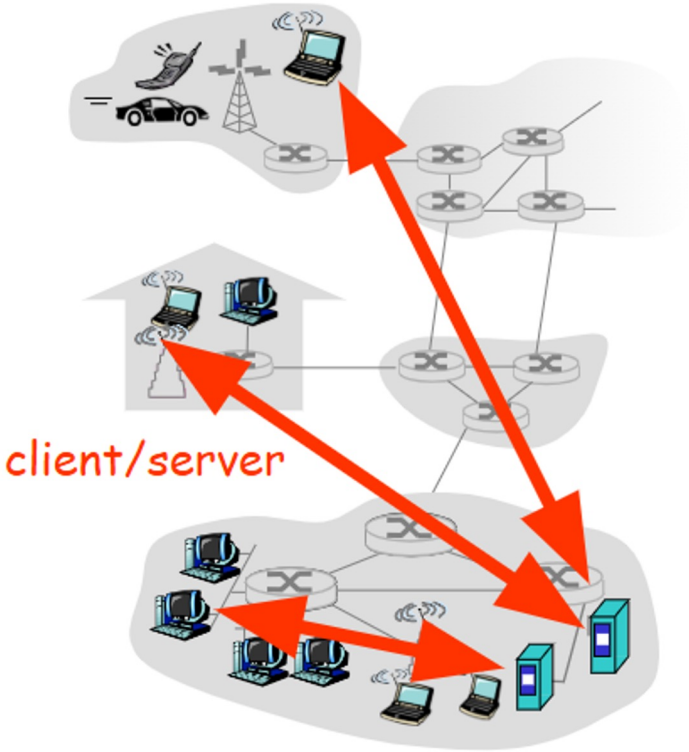
Highly scalable but difficult to manage

There's also a hybrid architecture that combines Client-Server and P2P

Client-Server vs. P2P

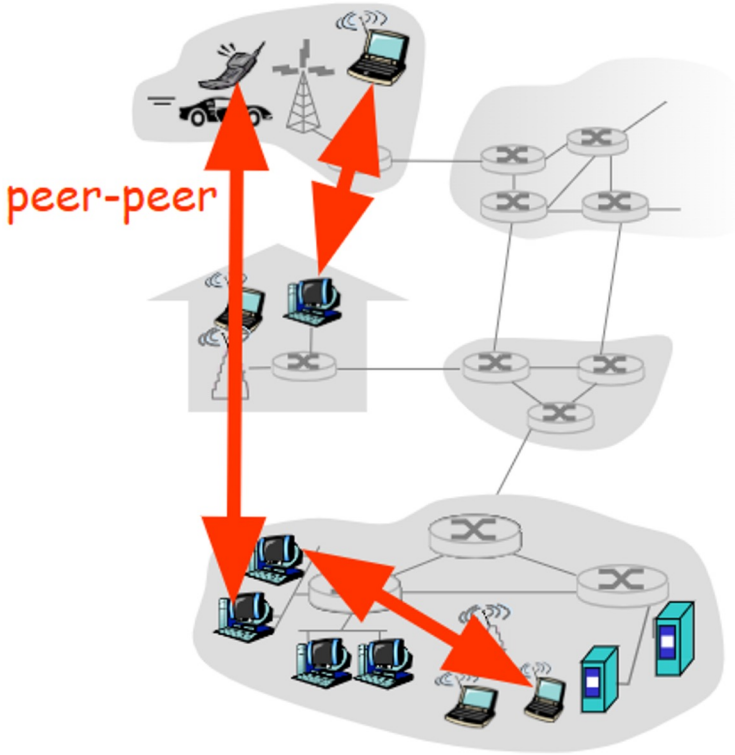
Client-Server

C-S



Peer-to-Peer

P2P



Processes Communicating

Process: program running within a host.

- Within the same host, two processes communicate using inter-process communication (defined by OS).
- Processes in different hosts communicate by exchanging messages

Client process and server process

- Client process: process that initiates communication
- Server process: process that waits to be contacted

App-layer protocol defines

Public-domain protocols:

- defined in RFCs
- allows for interoperability
- e.g., HTTP, SMTP

Proprietary protocols:

- e.g., Skype

01 **Types of messages exchanged,**

e.g., request, response

02 **Message syntax:**

what fields in messages & how fields are delineated

03 **Message semantics**

meaning of information in fields

03 **Rules for when and how processes send & respond to messages**

Internet apps: application, transport protocols

Application	Application layer protocol	Underlying transport protocol
e-mail	SMTP [RFC 2821]	TCP
remote terminal	Telnet [RFC 854]	TCP
access	HTTP [RFC 2616]	TCP
Web	FTP [RFC 959]	TCP
file transfer	HTTP (eg Youtube),	TCP or UDP
streaming multimedia	RTP [RFC 1889]	
Internet telephony	SIP, RTP, proprietary (e.g., Skype)	typically UDP

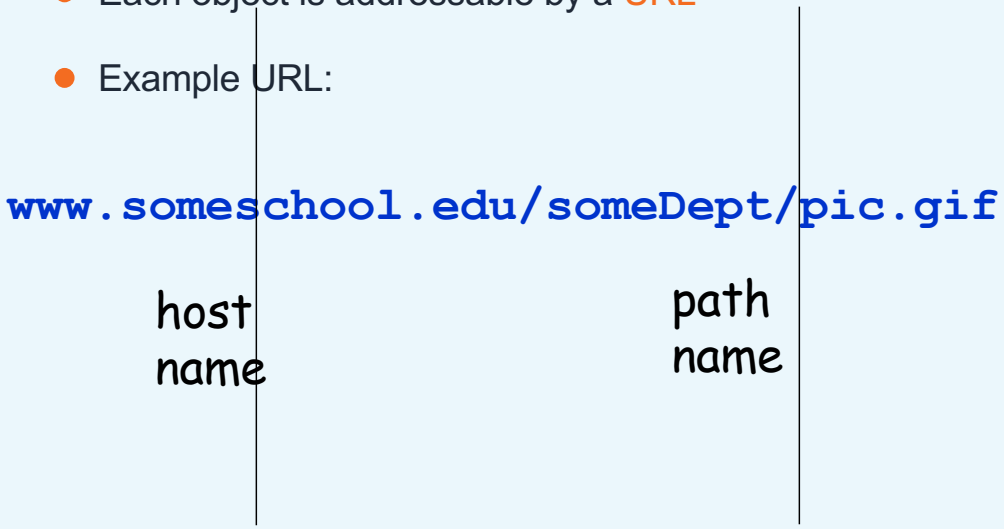
Web and HTTP

Web

- Web page consists of objects
- Object can be HTML file, JPEG image, Java applet, audio file,…
- Web page consists of base HTML-file which includes several referenced objects

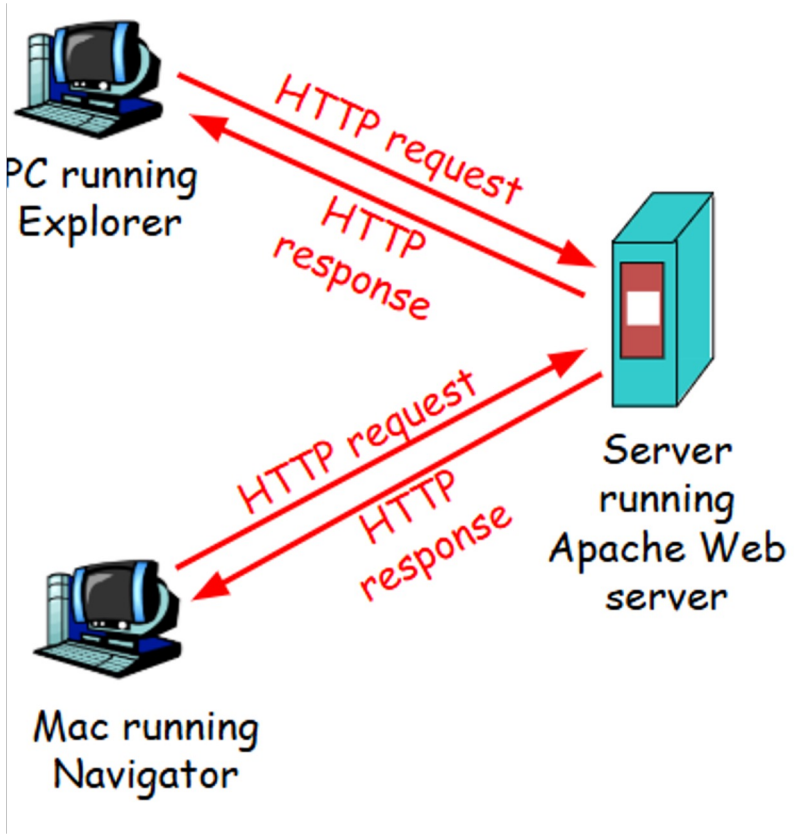
URL

- Each object is addressable by a URL
- Example URL:



HTTP

HyperText Transfer Protocol



01 **Web's application layer protocol**

02 **Client: browser that requests, receives, and "displays" Web objects**

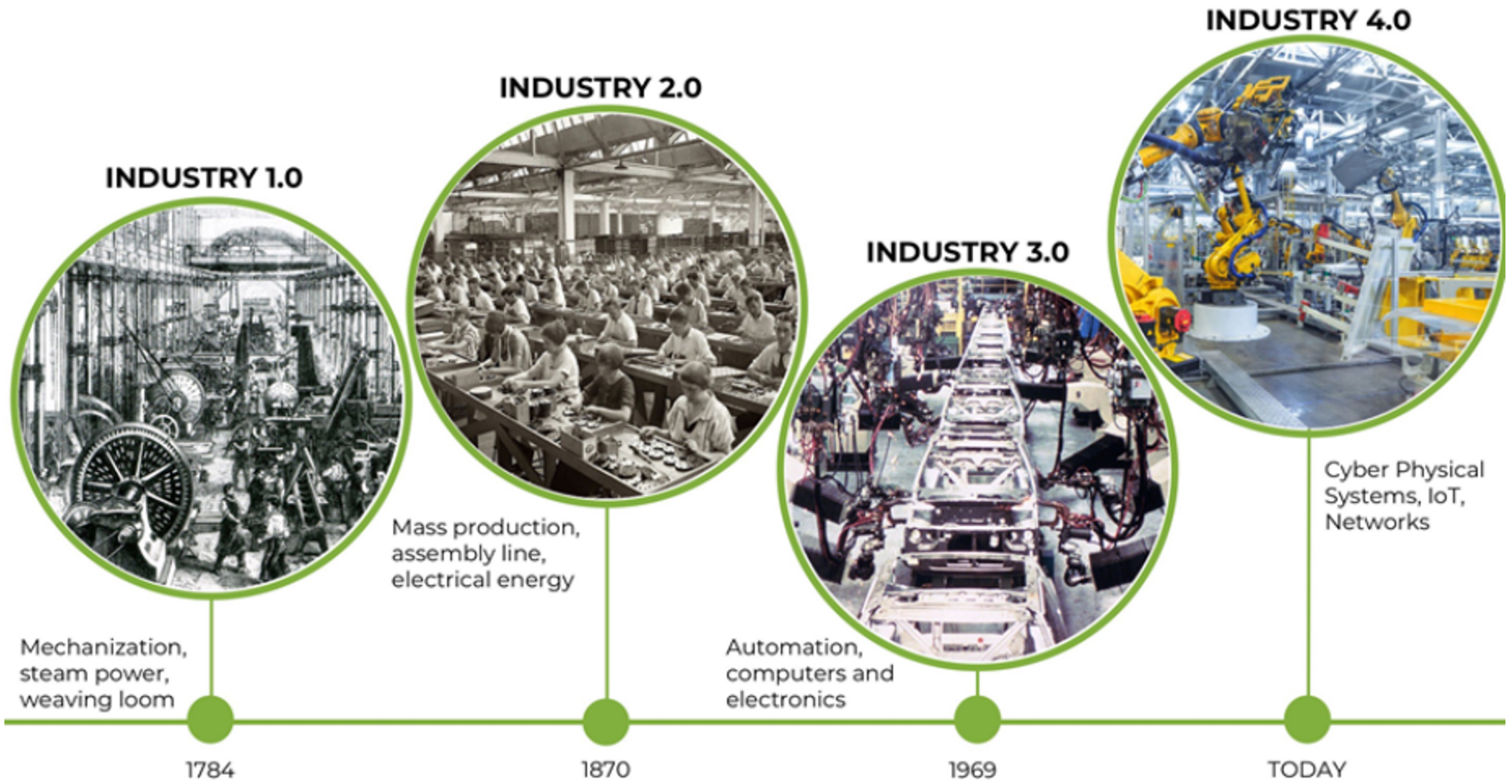
03 **Server: web server sends objects in response to requests.**

05

INTERNET OF THINGS (IOT) & EDGE AI

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Industry 4.0 and Internet of Things



Internet of Things (IoT) with AI

(Timothy Chou)

Do

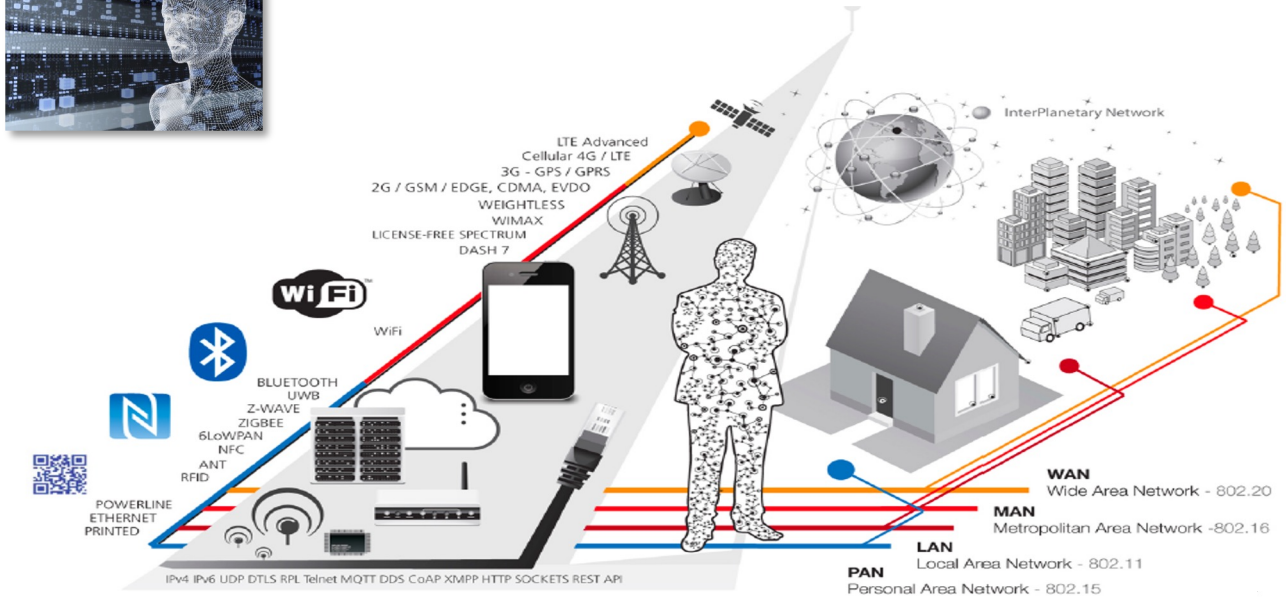




Learn

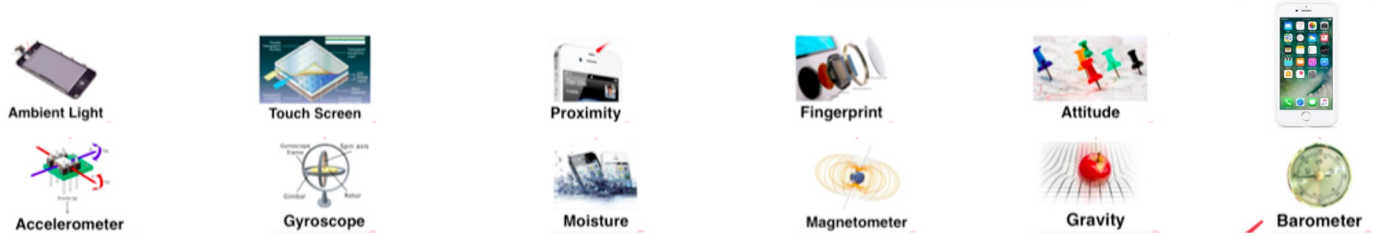


Collect

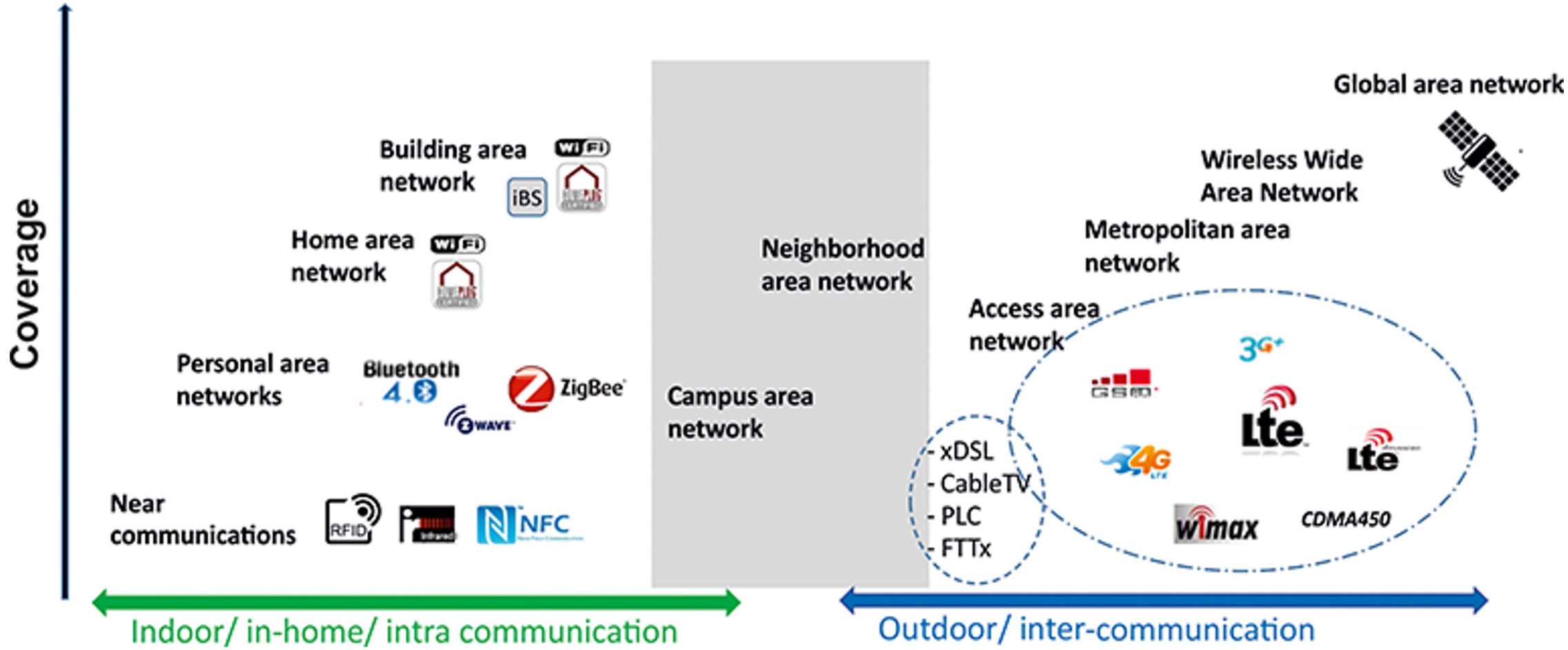


Connect

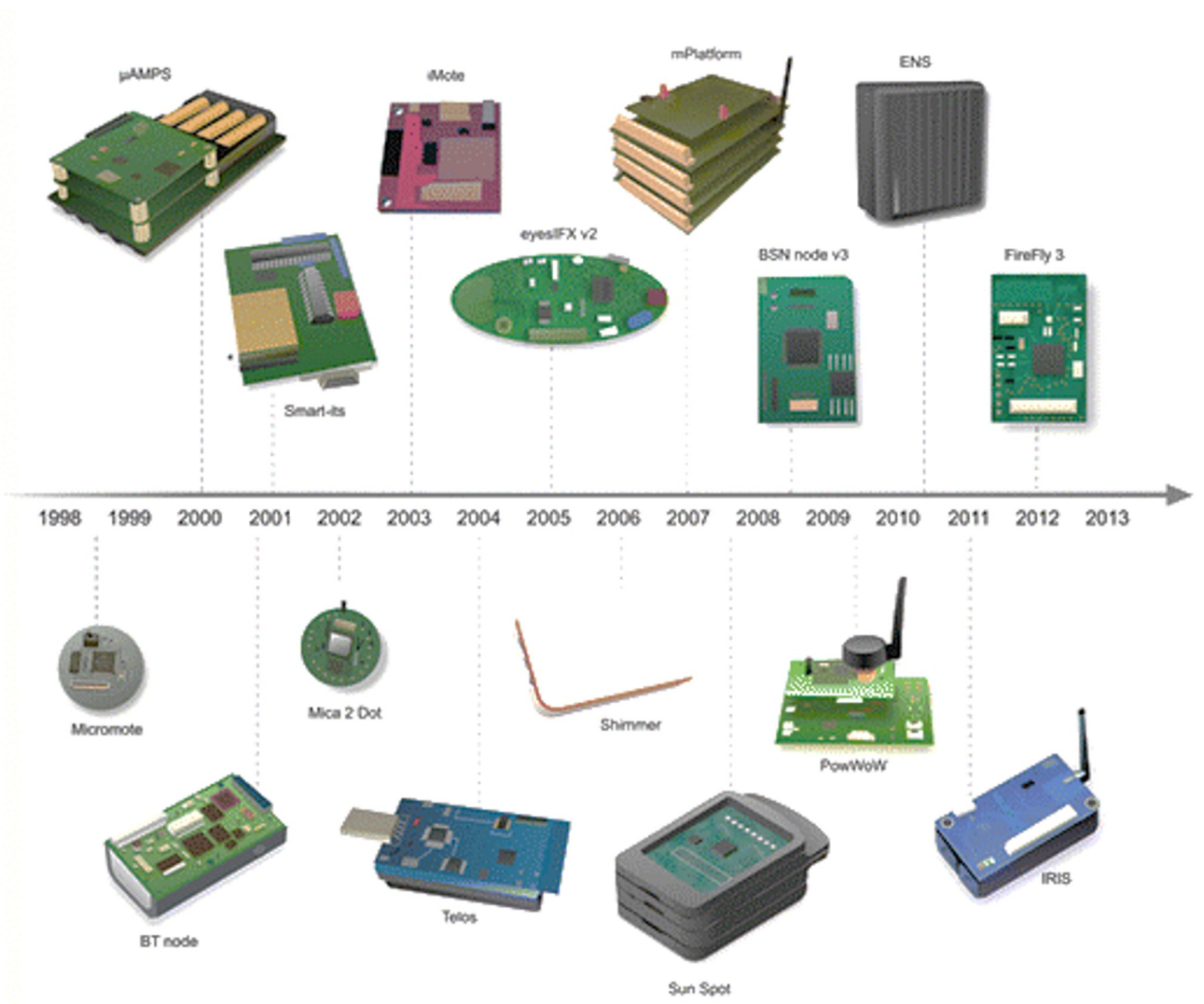
Things



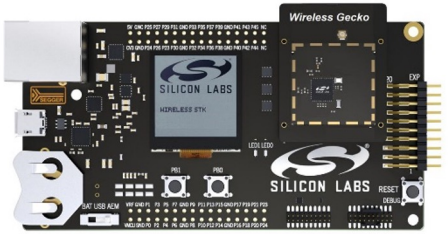
Connect Layer



IoT End-devices

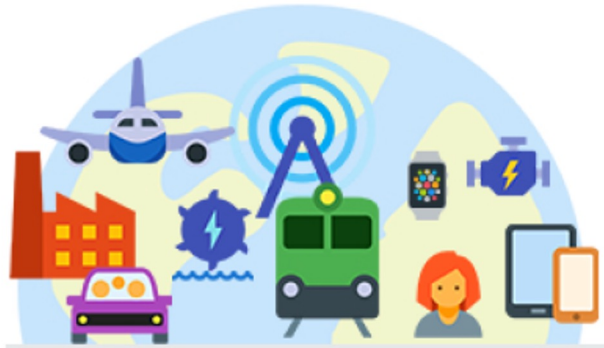








Edge Computing



Operations Technology

Information Technology

THE EDGE

The Things

- Wearables
- Cars
- Motors
- Buildings
- Mobile Devices
- Meters
- Robotics
- Generators

Sensors & Actuators

Data Aggregation & Gateways

Edge IT

Data Center & Cloud IT

DATA GENERATION

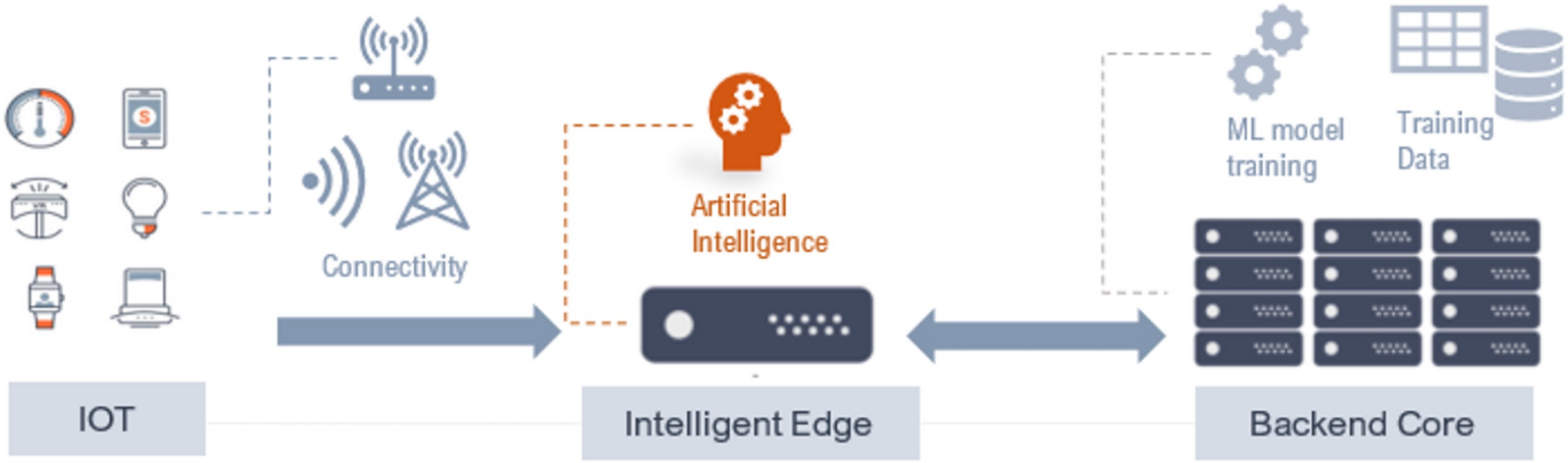
DATA SENSING

DATA COLLECTION AGGREGATION

EARLY DATA ANALYTICS

DEEP DATA ANALYTICS

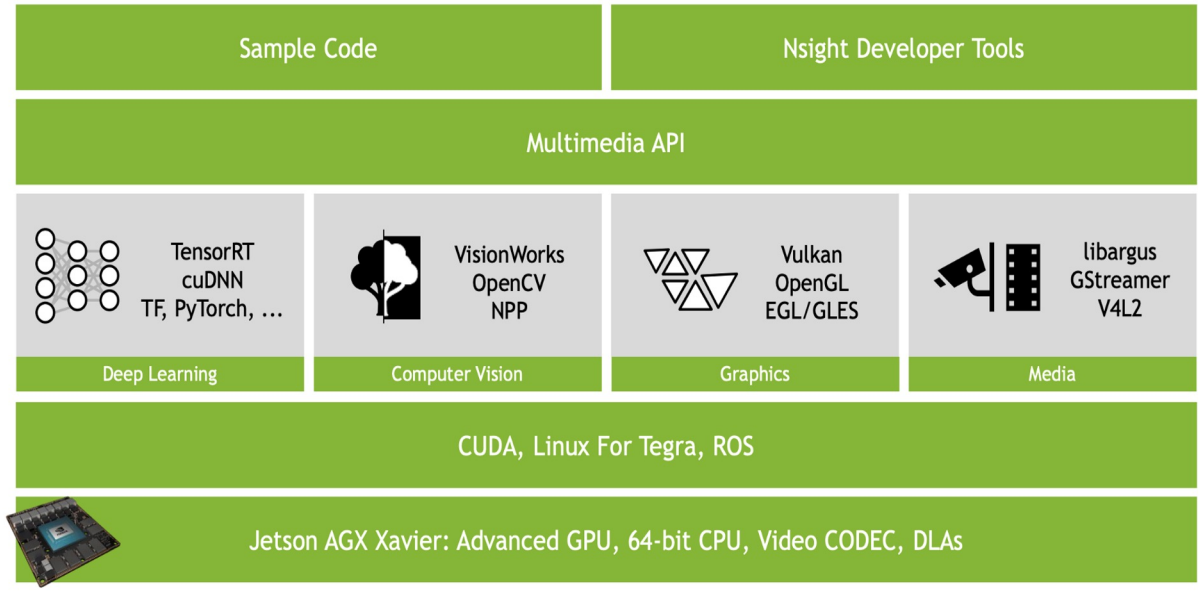
Combine it with AI = Edge AI



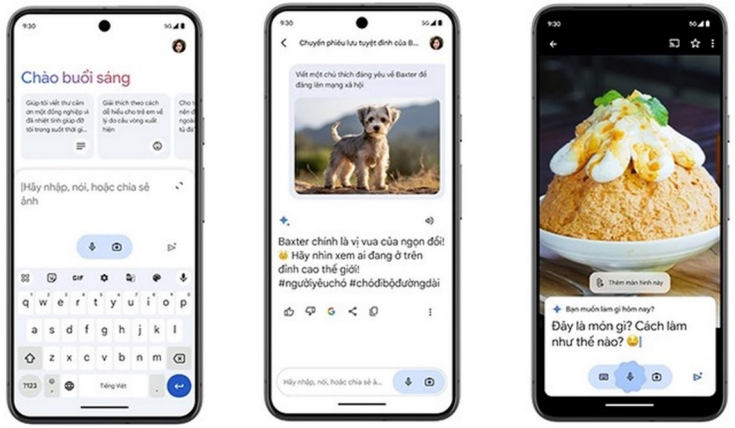
Edge AI overcomes the latency of data and complex computations:

- Increase in levels of Automation
- Digital Twins for Advanced Analytics
- Real-Time Decision Making
- Edge Inference and Training

Edge AI on Embedded Platform



Gemini

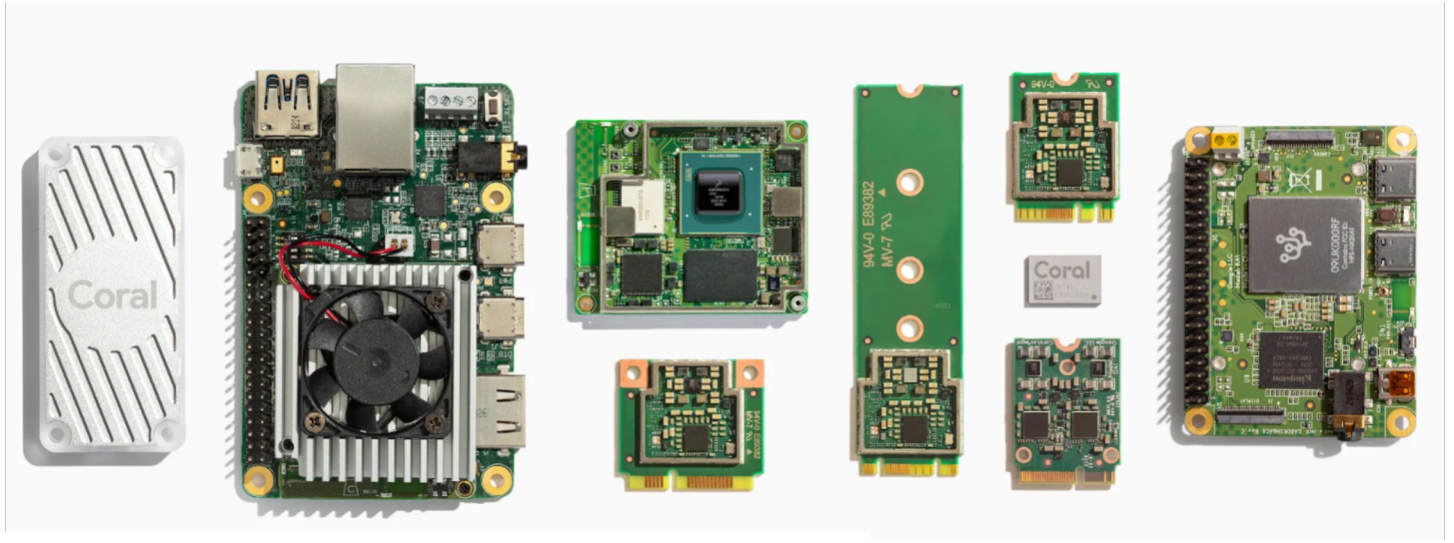


LLM on target device



384 NVIDIA CUDA cores and 48 Tensor Cores

PyCoral [https://coral.ai/products/]



Models

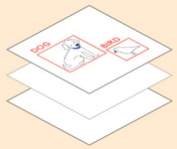
Trained TensorFlow models for the Edge TPU



Image classification

Models that recognize the subject in an image, plus classification models for on-device transfer learning.

→ See models



Object detection

Models that identify multiple objects and provide their location.

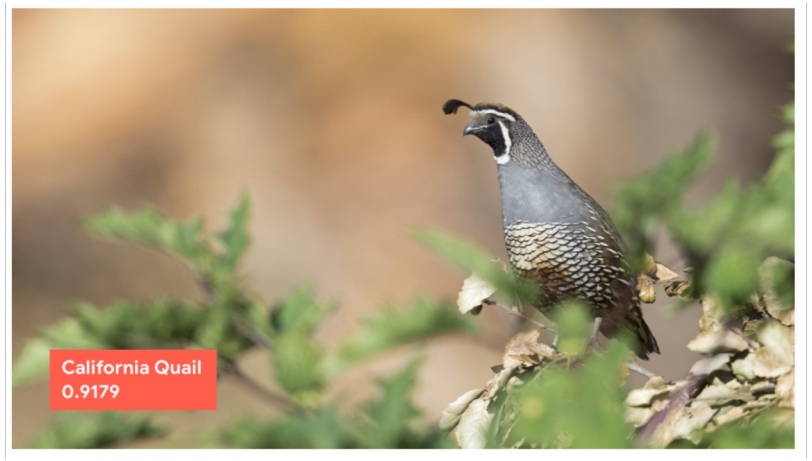
→ See models



Semantic segmentation

Models that identify specific pixels belonging to different objects.

→ See models

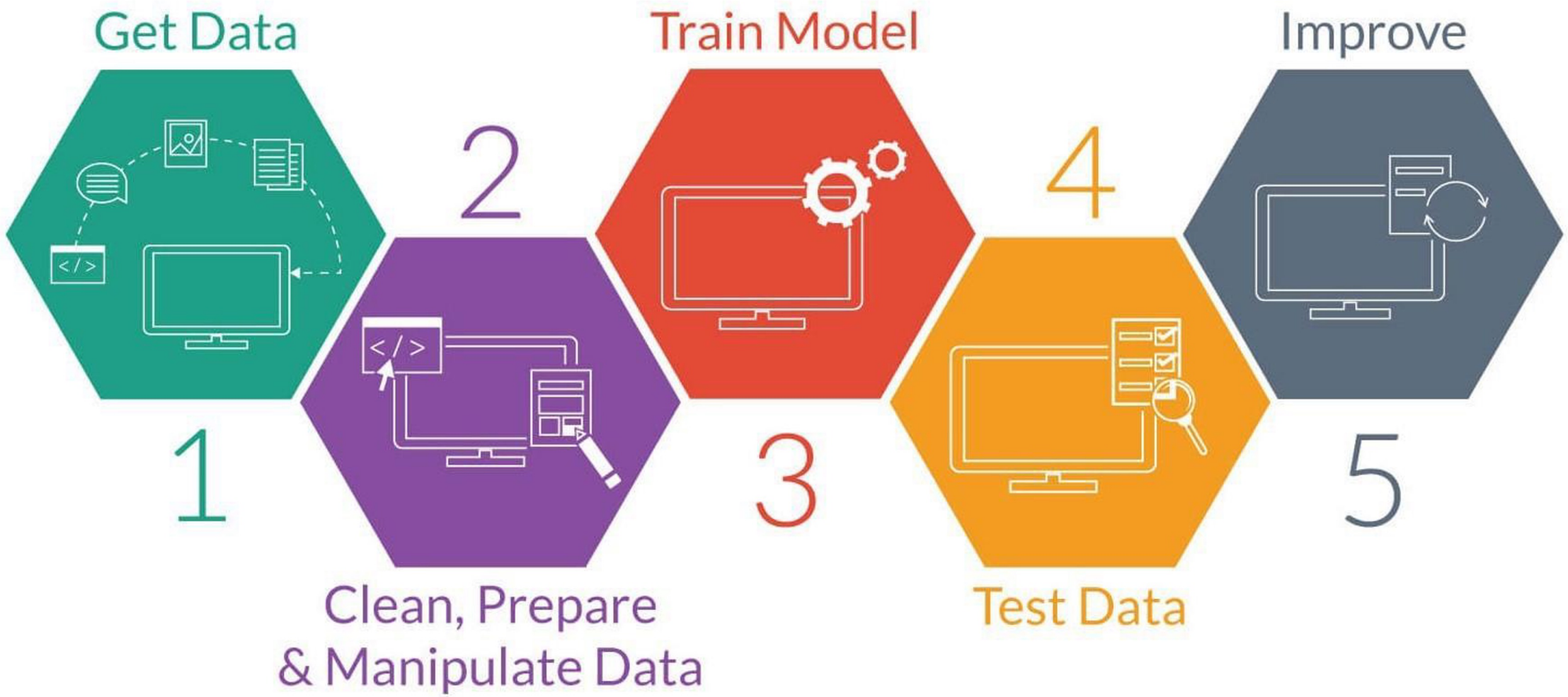


06

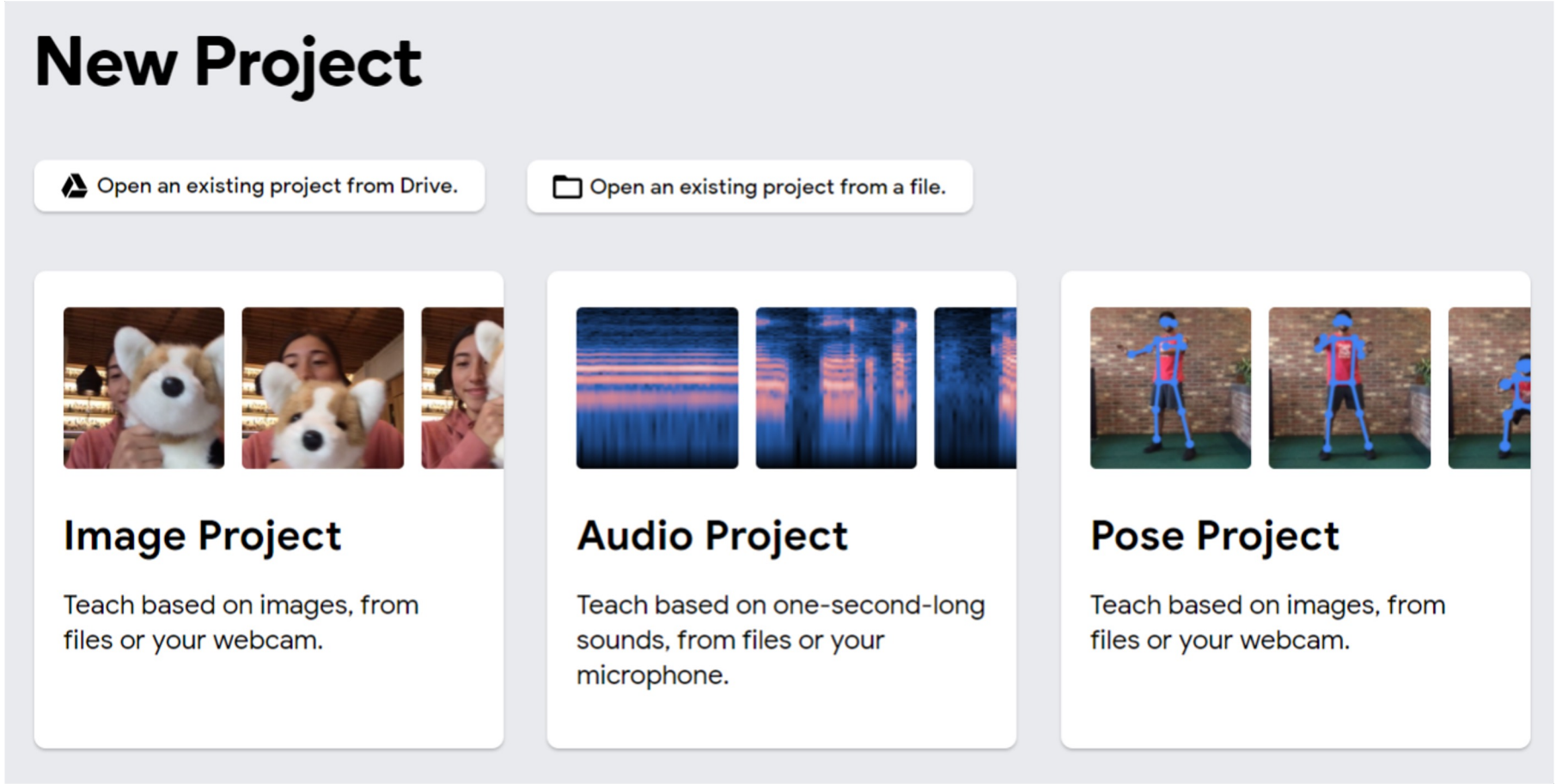
APPLICATIONS

Week [x] • CO1005 • [Class] • [Instructor]

Train an AI Model



Demo with Google TM [https://teachablemachine.withgoogle.com/train]



Smart Agriculture





Objectives

- Temperature and Humidity
- pH, EC
- Light intensity
- Light intensity

Monitoring system based on AR/VR



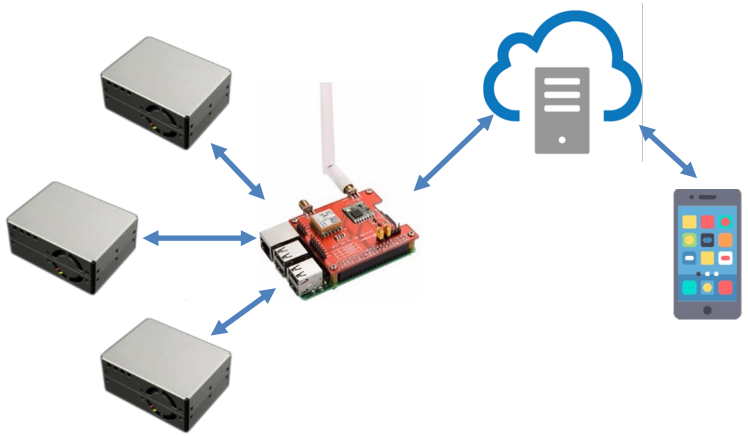






Air Quality Monitoring

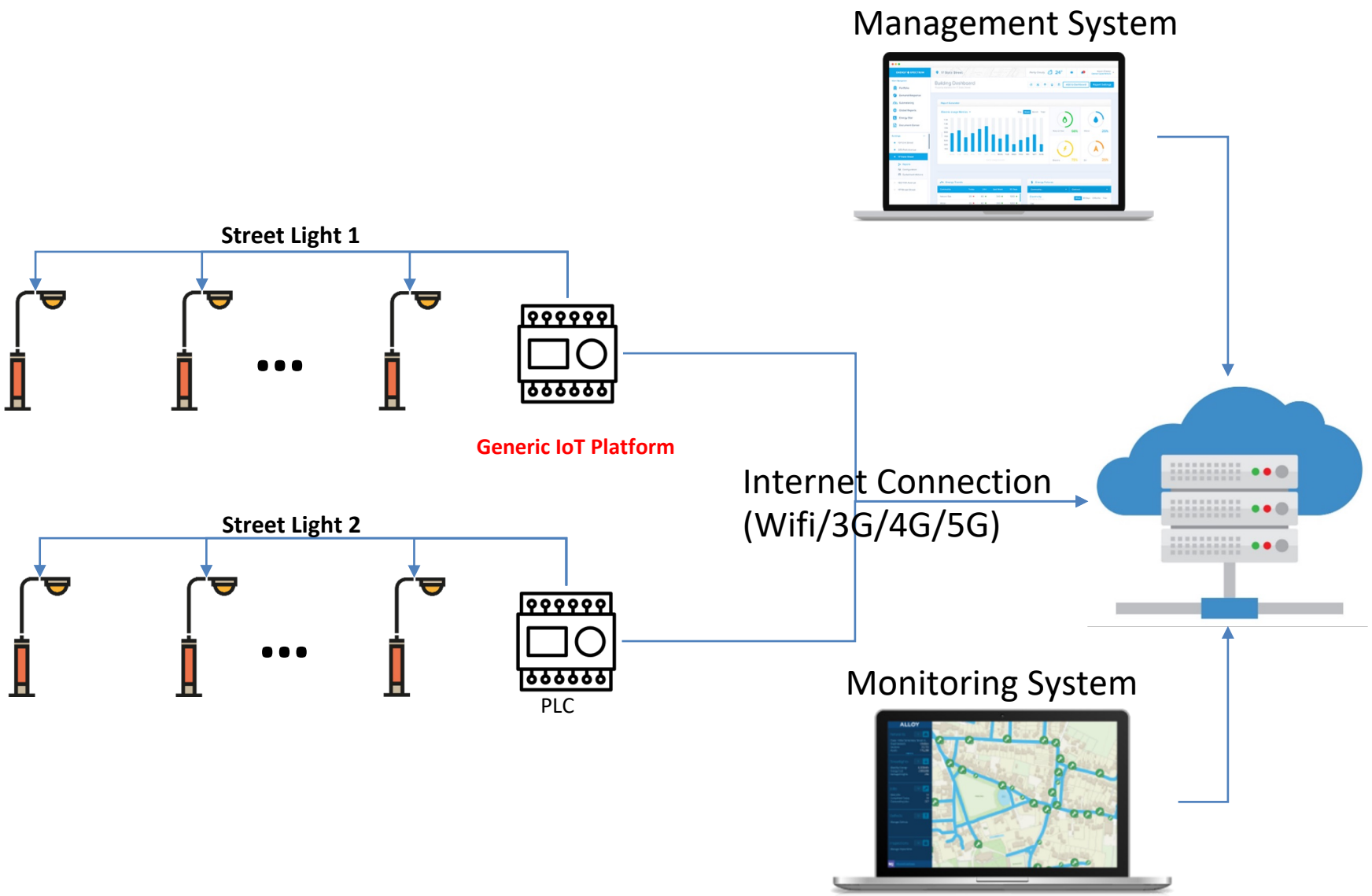




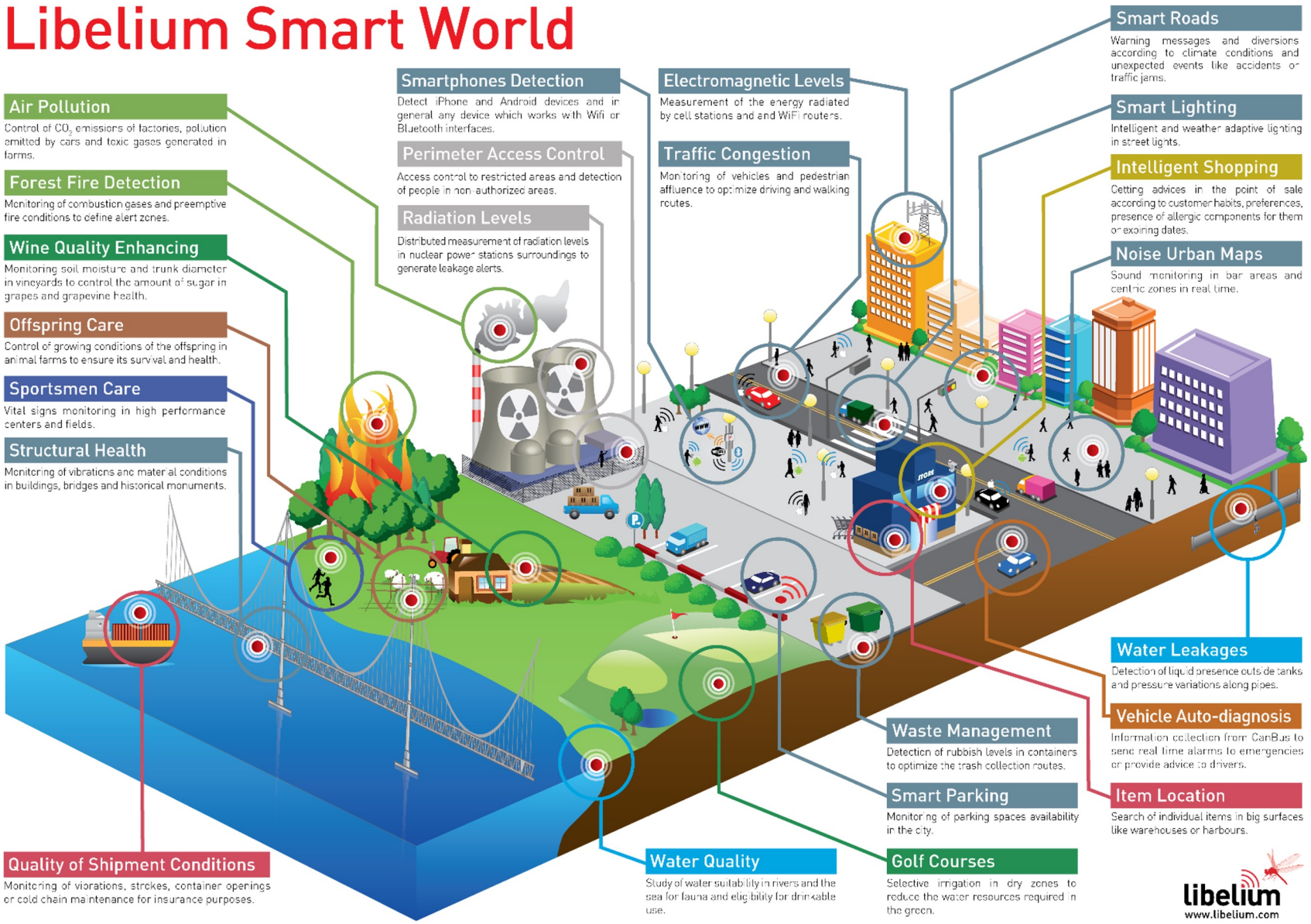
Objectives

- Air quality monitoring program assists us in improving and developing air pollution control programs to reduce the effect of air pollution.
- PM2.5, PM10, CO2, CO

Smart Street Light



Applications: Internet of Things



Key Takeaways

- **Foundation:** OSI & TCP/IP reference models and physical-to-wireless communication channels.
- **Mechanism:** Hierarchical network devices (Hub/Switch/Router), protocols, and edge computing.
- **Applications:** Real-world IoT deployments in smart agriculture, environmental monitoring, and intelligent city systems.
- **Next steps:** Preparing for chapter 3 (Introduction to Database)

Thank you for listening!